PV102 Day 3 Morning Quiz: Answer Key

10 questions | 15 minutes | Independent work

PV102 | CTS Day 19 | Morning Quiz - Answer Key
Watts are moments. Watt-hours are work over time. | Wednesday, August 5, 2026

Directions: Answer from your current knowledge. For written responses, use complete technical statements. If unsure, state the safe next action rather than guessing.

1. A 60 W load runs for 3 hours. How much energy is used?

- A. 20 Wh
- B. 60 Wh
- C. 180 Wh
- D. 3 Wh

Answer: C

Rationale/look-for: Energy = watts x hours = 60 x 3 = 180 Wh.

2. Watts measure:

- A. Instant power
- B. Total daily energy only
- C. Wire length
- D. Battery age

Answer: A

Rationale/look-for: Watts are a rate of using or producing energy.

3. Watt-hours measure:

- A. Energy over time
- B. Compass direction
- C. Module frame size
- D. Fuse color

Answer: A

Rationale/look-for: Watt-hours combine power and runtime.

4. The load that often changes inverter selection is:

- A. Sticky note
- B. Surge or motor load
- C. Pencil
- D. Label maker

Answer: B

Rationale/look-for: Starting surge can exceed running watts and affect inverter requirements.

5. A good client load interview asks:

- A. What must run, when, and for how long?
- B. What color is the inverter?
- C. How fast can we buy parts?
- D. Which battery looks best?

Answer: A

Rationale/look-for: Load definition begins with required function and runtime.

6. Calculate: 12 V x 5 A = ___ W.

_____

_____

Answer: 60 W

Rationale/look-for: Power = volts x amps.

7. Why should optional loads be separated from essential loads?

_____

_____

Answer: Priority

Rationale/look-for: Optional loads can be shed to reduce system size, cost, and storage requirements.

8. Name one load unknown that must be verified before final sizing.

_____

_____

Answer: Instructor judgment

Rationale/look-for: Accept actual watts, surge current, duty cycle, runtime, voltage, season, or user behavior.

9. What happens to required battery capacity when runtime doubles?

_____

_____

Answer: It roughly doubles

Rationale/look-for: If other factors stay constant, more runtime requires more stored energy.

10. Write the Day 3 rule in your own words.

Answer: Power vs energy

Rationale/look-for: Look for distinction between watts and watt-hours.